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COOPERATIVE RANDOM ACCESS FOR CELL-FREE NETWORKS

Abstract

This disclosure relates to cell-free network. A network device is provided, which maintains a first dataset indicating neighboring network devices, and second datasets each indicating one or more user IDs. One second dataset is associated with the network device, and each other second dataset is associated with one neighboring network device of the first dataset. The network device receives signals from user devices, wherein each signal comprises a respective user ID and a respective payload of one user device. The network device separates one or more signals to obtain the respective user IDs and payloads, and sends first exchange information to each neighboring network device that is in the first dataset and is associated with a second dataset including at least one obtained respective user ID. The sent first exchange information comprises at least one obtained respective user ID and respective payload.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/EP2022/082402, filed on Nov. 18, 2022, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
[0002] The present disclosure relates to communication networks and systems, specifically cell-free communication networks. The disclosure provides a network device for a cell-free wireless communication network, and a method for operating the network device.
BACKGROUND
[0003] Cell-free networks are composed of multiple network devices, e.g., access points (APs), which serve multiple user devices (e.g., UEs), wherein the user devices are not attached to a particular network device.
[0004] Conventional cell-free networks operate based on the assumption of a multiple-access scenario, wherein active user devices are known, and wherein pilot sequences are assigned to these active user devices in a coordinated manner (i.e., conventional cell-free networks are not grant-free). Uplink (UL) communications of the user devices are then divided into two steps: a first step of channel estimation, wherein the pilot sequences are transmitted by the user devices; and a second step of communication, wherein the user devices transmit their messages. Random-access schemes in the conventional cell-free networks further perform a step of activity detection based on preamble sequences.
[0005] Notably, “grant-free” means that in the network, one or more transmitting devices (UEs for the UL) transmit messages to one or more receiving devices (APs for the UL) without any prior resource request or grant. Further, “random access” means that a random number of transmitting devices in the network are active at the same time. Grant-free and random access communications are typical for massive internet of things (IoT) scenarios.
[0006] An issue is, that for communication networks and systems, in which activity detection and decoding are to be performed jointly (cooperatively) by network devices, the above-described activity detection methods of the conventional cell-free networks are not applicable. Thus, new solutions are needed for cooperative random access for cell-free networks.
SUMMARY
[0007] In view of the above, the present disclosure is related to cell-free networks and random access. An objective of this disclosure is to enable cooperative random access for a cell-free (wireless) network. A particular objective is to enable cooperative decoding of signals transmitted by multiple user devices to multiple network devices in the cell-free network.
[0008] These and other objectives are achieved by this disclosure as described in the enclosed independent claims. Advantageous implementations are further defined in the dependent claims.
[0009] A first aspect of this disclosure provides a network device for a cell-free wireless communication network, the network device being configured to: maintain a first dataset indicating one or more neighboring network devices; maintain a plurality of second datasets each indicating one or more user IDs, wherein one second dataset of the plurality of second datasets is associated with the network device, and each other second dataset of the plurality of second datasets is

associated with one of the one or more neighboring network devices in the first dataset; receive a plurality of signals from a plurality of user devices in a first interval, wherein each signal comprises a respective user ID of one of the user devices and a respective payload of one of the user devices; separate, in the first interval, one or more signals of the plurality of signals to obtain the respective user IDs and the respective payloads of the one or more signals; and send, in the first interval, first exchange information to each neighboring network device, which is in the first dataset and is associated with a second dataset that includes at least one of the obtained respective user IDs; wherein the sent first exchange information comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.

[0010] By maintaining the first dataset and second datasets, and by sending the first exchange message to the neighboring network devices (which may also maintain a first dataset and second datasets), the network device of the first aspect supports cooperative separation (decoding) of the plurality of signals transmitted by the multiple user devices to the network device and the neighboring network devices. Thus, the network device of the first aspect supports cooperative random access in a cell-free network, for instance, a cell-free wireless/mobile network.

[0011] In an implementation form of the first aspect, if a particular user ID of the obtained respective user IDs is not in the second dataset associated with the network device, the network device is configured to send, in the first interval, second exchange information to each neighboring network device, which is in the first dataset and is associated with a second dataset that does not include the particular user ID; wherein the second exchange information comprises the particular user ID but not the respective payload corresponding to the particular user ID.

[0012] This allows decreasing the signaling overhead for the cooperative random access method.

[0013] In an implementation form of the first aspect, the network device is further configured to: receive, in the first interval, first exchange information from a neighboring network device, wherein the received first exchange information comprises at least one of the respective user IDs and at least one of the respective payloads of the plurality of signals; and further separate, in the first interval, one or more of the plurality of signals based on the separated signals and the respective payloads in the received first exchange information.

[0014] Accordingly, the network device may exchange first exchange message with the neighboring network devices to enable the cooperative separation (decoding) of the plurality of signals transmitted by the multiple users.

[0015] In an implementation form of the first aspect, the network device is further configured to send, in a second interval, the second exchange information to each neighboring network device in the first dataset.

[0016] In an implementation form of the first aspect, the network device is further configured to perform the separation of the one or more signals of the plurality of signals, in the first interval, without prior knowledge of the respective user IDs and the respective payloads of the one or more signals.

[0017] Thus, random access in a cell-free network is supported.

[0018] In an implementation form of the first aspect, the first exchange information received from the neighboring network device comprises one or more respective user IDs and one or more respective payloads that are different than in the sent first exchange information.

[0019] In an implementation form of the first aspect, in the further separation of the one or more signals of the plurality of signals, in the first interval, the network device is configured to perform interference cancellation.

[0020] Thus, an improved separation of the plurality of signals is achieved.

[0021] In an implementation form of the first aspect, the network device is further configured to update the second dataset associated with the network device based on the one or more separated signals and the one or more further separated signals, and/or update one or more second datasets associated with one or more neighboring network devices based on first exchange information received from the one or more neighboring network devices.

[0022] Accordingly, the network device of the first aspect (and in a similar manner this may also be done by the neighboring network devices), maintains up-to-date information on the connections in the cell-free network.

[0023] In an implementation form of the first aspect, the network device is configured to update the second dataset associated with the network device and/or the one or more second datasets associated with the one or more neighboring network devices by at least one of: adding one or more user IDs to the second dataset associated with the network device; and adding one or more user IDs to one or more second datasets associated with one or more neighboring network devices; and removing one or more user IDs from the second dataset associated with the network device; and removing one or more user IDs from the one or more second datasets associated with the one or more neighboring network devices.

[0024] In an implementation form of the first aspect, the network device is further configured to associate a timestamp with any user ID added to a second dataset.

[0025] In an implementation form of the first aspect, the network device is further configured to remove any user ID from a second dataset, if a timestamp associated with the user ID is older than a threshold.

[0026] In an implementation form of the first aspect, the network device is further configured to adjust the threshold based on at least one of a network condition and an upper layer configuration.

[0027] In an implementation form of the first aspect, the network device is further configured to: add a user ID of a separated signal or a further separated signal to the second dataset of the network device, if the user ID of the separated signal is not already in the second dataset of the network device; and/or add a user ID in first exchange information received from a neighboring network device to the second dataset of that neighboring network device, if the user ID in the first exchange information is not already in the second dataset of that neighboring network device.

[0028] A second aspect of this disclosure provides a method for a network device of a cell-free wireless communication network, the method comprising: maintaining a first dataset indicating one or more neighboring network devices; maintaining a plurality of second datasets each indicating one or more user IDs, wherein one second dataset of the plurality of second datasets is associated with the network device, and each other second dataset of the plurality of second datasets is associated with one of the one or more neighboring network devices in the first dataset; receiving a plurality of signals from a plurality of user devices in a first interval, wherein each signal comprises a respective user ID of one of the user devices and a respective payload of one of the user devices; separating, in the first interval, one or more signals of the plurality of signals to obtain the respective user IDs and the respective payloads of the one or more signals; and sending, in the first interval, first exchange information to each neighboring network device, which is in the first dataset and is associated with a second dataset that includes at least one of the obtained respective user IDs; wherein the first exchange information comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.

[0029] The method of the second aspect may have implementation forms, which correspond to the implementation forms of the network device of the first aspect. The method of the second aspect and its possible implementation forms achieve the same advantages as described above for the network device of the first aspect and its respective implementation forms.

[0030] A third aspect of this disclosure provides a computer program comprising instructions which, when the program is executed by a computer, cause the computer to perform the method according to the second aspect and its possible implementation forms.

[0031] A fourth aspect of this disclosure provides a non-transitory storage medium storing executable program code which, when executed by a processor, causes the method according to the second aspect and its possible implementation forms to be performed.

[0032] According to the above aspects and implementation forms, this disclosure addresses the cooperative separation of signals transmitted by multiple user devices (e.g., multiple UEs) to multiple network devices (e.g., multiple APs) in a cell-free wireless network.

[0033] The disclosure proposes the first dataset, the contents of which may be represented by a connectivity graph indicating the knowledge of the connections between the user devices and the network devices. Each network device may maintain such a first dataset. The first datasets may be employed to exchange signals (first and second exchange information) between the network devices sharing the same user devices. These exchange signals may then be used by the network devices to perform interference cancelation, thus improving the decoding performance. The way to build up and use the first datasets in a distributed manner is explained in this disclosure. In particular, the knowledge of the first dataset at each network device may be reduced to the knowledge of the second datasets. The content of the second datasets which may be represented by a neighboring graph including first and second order neighboring network devices of a respective network device.

[0034] It has to be noted that all devices, elements, units and means described in the present application could be implemented in the software or hardware elements or any kind of combination thereof. All steps which are performed by the various entities described in the present application as well as the functionalities described to be performed by the various entities are intended to mean that the respective entity is adapted to or configured to perform the respective steps and functionalities. Even if, in the following description of specific embodiments, a specific functionality or step to be performed by external entities is not reflected in the description of a specific detailed element of that entity which performs that specific step or functionality, it should be clear for a skilled person that these methods and functionalities can be implemented in respective software or hardware elements, or any kind of combination thereof.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0035] The above-described aspects and implementation forms will be explained in the following description of specific embodiments in relation to the enclosed drawings, in which FIG. **1** shows a network device for a cell-free network, according to this

DISCLOSURE

[0036] FIG. **2** shows (a) an exemplary connectivity graph, and (b) an exemplary neighboring graph.

[0037] FIG. **3** illustrates (a) information encoding at a user device, and (b) an example of an encoder of a user device.

[0038] FIG. **4** shows a decoding timeline for two consecutive intervals, wherein the decoding is performed by a network device according to this disclosure.

[0039] FIG. **5** shows (a) a separation of signals of multiple user devices in a first decoding step, and (b) an example for information decoding at a network device.

[0040] FIG. **6** shows an exemplary scheme of exchanging information for each decoded user device between neighboring network devices.

[0041] FIG. **7** shows a further separation of signals of multiple user devices in a second decoding step.

[0042] FIG. **8** shows an illustrative example of a decoding scheme proposed by this disclosure with four intervals.

[0043] FIG. **9** shows a first interval of the example.

[0044] FIG. **10** shows a second interval of the example.

[0045] FIG. **11** shows a third interval of the example.

[0046] FIG. **12** shows a fourth interval of the example.

[0047] FIG. **13** shows a method for a network device of a cell-free network, according to this disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0048] FIG. **1** shows a network device **100** for a cell-free wireless communication network. The communication network may be a mobile communication network. For instance, the mobile communication network may be a 5.sup.th generation (5G), new radio (NR), or other next generation mobile network. The network device may be an AP, or a base station (BS), or a gNodeB, or another type of network access device.

[0049] The network device **100** is configured to maintain a first dataset **101**, which indicates one or more neighboring network devices **100** of the network device **100**. The neighboring network devices **100** could be first-order neighboring network devices **100** of the network device **100**, or second-order neighboring network devices **100**. Notably, each neighboring network device **100** may be configured like the discussed network device **100**.

[0050] The network device **100** is also configured to maintain a plurality of second datasets **102**. Each second dataset **102** indicates one or more user IDs. One of the second datasets **102** is associated with the network device **100**, and each other one of the second datasets **102** is associated with one of the one or more neighboring network devices **100** that are indicated in the first dataset **101**.

[0051] The network device **100** is further configured to receive a plurality of signals **103** from a plurality of user devices **110** in a first interval. Each user device **110** may be a UE or terminal device. Each signal of the plurality of signals **103** comprises a respective user identification (ID) of one of the user devices **110** and comprises a respective payload (message) of one of the user devices **110**. Notably, the plurality of signals **103** from the user devices **110** may also be received by the one or more neighboring network devices **100**. That is, a neighboring network device **100** of the network device **100** may be one that can potentially receive and decode a user device signal that the network device **100** can also receive and decode.

[0052] The network device **100** is configured to separate, in the first interval, one or more signals of the plurality of signals **103**, in order to obtain the respective user IDs and the respective payloads of these one or more signals. That is, for each separated signal, the network device **100** may obtain the respective user ID and the respective payload.

[0053] Then, the network device **100** is configured to send, in the first interval, first exchange information **104** to each neighboring network device **100**, which is indicated in the first dataset **101** and which is associated with a second dataset **102** that includes at least one of the obtained respective user IDs. The sent first exchange information **104** comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.

[0054] Moreover, the network device **100** may receive, also in the first interval, similar first exchange information **104** from at least one of the neighboring network devices **100**. The received first exchange information **104** comprises at least one of the respective user IDs and at least one of the respective payloads of the plurality of signals **103**. The first exchange information **104** received from the at least one neighboring network device **100** may comprise one or more respective user IDs and one or more respective payloads that are different than those in the sent first exchange information **104**. Accordingly, the network device **100** and its neighboring network devices **100** may exchange first exchange information **104**.

[0055] The network device **100** may additionally be configured to further separate, in the first interval, one or more of the plurality of signals **103** based on the (previously) separated signals and based on the respective payloads included in the received first exchange information **104**. In this sense, the first exchange information **104** is used for further separating the signals, and this may be done at each network device **100**. The network device **100** and the neighboring network devices **100** may thus collaborate to separate the signals.

[0056] Notably, the separating and the further separating of the signals of the plurality of signals **103** may be considered decoding steps. Thus, the network device **100** may participate in cooperative decoding of the signals transmitted by the multiple user devices (to the network device **100** and the neighboring network devices **100**), specifically in a cell-free wireless network.

[0057] Each network device **100** may comprise a processor or processing circuitry (not shown) configured to perform, conduct or initiate the various operations of the network device **100** described herein. The processing circuitry may comprise hardware and/or the processing circuitry may be controlled by software. The hardware may comprise analog circuitry or digital circuitry, or both analog and digital circuitry. The digital circuitry may comprise components such as application-specific integrated circuits (ASICs), field-programmable arrays (FPGAs), digital signal processors (DSPs), or multi-purpose processors. Each network device **100** may further comprise memory circuitry, which stores one or more instruction(s) that can be executed by the processor or by the processing circuitry, in particular under control of the software. For instance, the memory circuitry may comprise a non-transitory storage medium storing executable software code which, when executed by the processor or the processing circuitry, causes the various operations of the network device **100** to be performed. In one embodiment, the processing circuitry comprises one or more processors and a non-transitory memory connected to the one or more processors. The non-transitory memory may carry executable program code which, when executed by the one or more processors, causes the network device **100** to perform, conduct or initiate the operations or methods described herein.

[0058] Further considerations and details of the above-discussed solutions of this disclosure are described in the following.

[0059] Within an area of interest, the communication network or system may comprise a set of network devices **100** (e.g., APs), including the network device **100** and its neighboring network devices **100** discussed above, and optionally further network devices, and may comprise a set of user devices **110** (e.g., UEs). For representing such a communication network or system, the first dataset **101** and a plurality of second datasets **102** may be used, for instance, individually maintained at each network device **100**. The content of these datasets **101**, **102** may be represented by two graphs, a connectivity graph as shown in FIG. **2**(*a*), and a neighboring graph as shown in FIG. **2**(*b*). These graphs may particularly illustrate the contents of all datasets **101**, **102**.

[0060] The connectivity graph may have two sorts of nodes, the network devices **100** (for example APs, as used in the figures) and the user devices **110** (for example UEs, as used in the figures). It is a bipartite graph to describe the connectivity between the user devices **110** and the network devices **100**. This means that a network device node can only be connected to one or more user device nodes of the graph. Similarly, a user device node can only be connected to one or more network device nodes of the graph. An edge in the connectivity graph between a network device **100** and a user device **110** indicates that a message transmitted by the user device **110** has been decoded by the network device **100** in the recent past.

[0061] The neighboring graph has only one kind of node, namely network devices **100**. This graph, i.e., the data it represents, may be known a priori by the network devices **100**. Two network devices **100** may be connected in the neighboring graph, if they can potentially decode a common user device message. The following example criteria can be used to determine, if two network devices **100** are connected in the neighboring graph: (i) propagation characteristics ensure sufficient receive signal strength at both network devices **100**; (ii) geometrical consideration of the geographical scene such as the distance between the two network devices **100** and a user device **110** is below some threshold; (c) the distance between the two network devices **100** is below some threshold (d) backhaul network architecture characteristics.

[0062] Notably, the illustration of the neighboring graph is an overview of the set of network devices **100**. In contrast, each network device **100** may have only partial knowledge of the graph in the sense that each network device **100** may know only its neighboring network devices **100** (that is, individual second datasets may be maintained by the network devices **100**).

[0063] The two graphs in FIG. **2** are for illustrative purposes only. The knowledge of these graphs may be acquired through two objects that each network device **100** stores and maintains locally: (i) a list of neighboring network devices **100** (an example of the first dataset **101**); (ii) for each network device **100** in the list of neighboring network devices **100** and the local network device **100** itself, a list of user devices **110** connected to the network device **100** (an example of the second datasets **102**), wherein any list can be empty if the considered network device **100** has no connected user device **110**. The disclosure also refers to these lists as lists of user IDs or lists of UIDs. Each element of the lists of user IDs may be tagged by a timestamp. In this disclosure, UL communication is considered as an example, where user

[0064] devices **110** and network devices **100** are the transmitters and receivers, respectively. This UL communication happens during multiple time intervals. During each interval, a subset of the user devices **110** (denoted as active user devices **110**) may transmit a payload (message) consisting of a sequence of bits. The fact that a user device **110** is active or inactive is not known by the network devices **100**. Since a cell-free network is configured in this disclosure, each user device **110** is not attached to a particular network device **100**, and its payload can be decoded by any network device **100**.

[0065] Taking the example of a single-input multiple-output block fading multiple access channel (SIMO-BF-MAC), one can assume that, for each interval, the user devices **110** and network devices **100** are synchronized in time, so that the signal received by a network device **100** is the sum of the contribution of each user device **110**. With these assumptions, at a particular network device l (wherein the index l may refer to the identifier of the network device **100**) of Nl receiving antennas, during an interval of length T, the baseband received signal Y.sub.l∈custom-character .sup.T×N.sup.l from K.sub.l active user devices **110** can be expressed as

[00001] $Y_l = \sum_{k=1}^{K_l} s_k h_{k,l}^T + W_l$ [0066] where s.sub.k∈custom-character.sup.T is the transmitted signal of the user device k, h.sub.k,l∈custom-character.sup.N.sup.l is the channel vector between the user device k and the network device l, and W.sub.lcustom-character .sup.T×N.sup.l is an additive noise term. The signal Y.sub.l may accordingly be the plurality of signals **103** referred to above.

[0067] Each user device **110** may encode its signal to a vector symbol s.sub.k chosen in the common vector constellation C. The vector constellation can be characterized as a set of 2.sup.B complex vectors

C={*c*.sub.1, . . . ,*c*.sub.2.sub.B}.

[0068] Note that this constellation may be common to all user devices **110**, and may be known by all the user devices **110** and the network devices **100**. The signal transmitted in each interval may be composed of the user ID and payload bits. FIG. **3**(*a*) illustrates an example of a transmission scheme that may be used in this disclosure

[0069] An example of the encoder is shown in FIG. **3**(*b*), and may be configured as follows. The user ID and the payload bits may be concatenated, and this concatenated sequence of bits may be translated into an integer index corresponding to an element of the constellation C with 2.sup.B.sup.1.sup.+B.sup.2 elements (hence, B=B.sub.1+B.sub.2).

[0070] The network device **100** proposed by this disclosure processes the signals sent by user devices **110** following four steps that are now detailed below: (i) a first decoding step, (ii) an exchange step, (iii) a second decoding step, and (iv) an update step. FIG. **4** shows the decoding steps for two successive intervals, namely the first interval **401** and the second interval **402**.

[0071] The steps (i), (iii), and (iv) may be performed independently at each network device **100**, e.g., based on the current knowledge of the local lists of UIDs, i.e., the examples of the second datasets **102** maintained by that network device **100**. These local lists of UIDs may evolve over time during each step (iv). The processing described in this disclosure is distributed, since no central network device or node with full knowledge of the transmitted signals is involved in the processing, nor is a full connectivity graph of the network used.

[0072] In the first decoding step, each network device **100** separates received signals **503** of the plurality of signals **103** of the plurality of user devices **110** without prior knowledge about either the user ID or payload. FIG. **5**(*a*) depicts the input and output for this step, which may be done by a receiver **501** of the network device **100**, and where ŝ.sub.k∈custom-character.sup.T is a recovered signal **503** of the user device k and {circumflex over (K)}.sub.l denotes the number of active user devices decoded by the network device l.

[0073] The user ID and the payload of each separated signal **503** can be recovered through a constellation demapper **502** as shown in FIG. **5**(*b*).

[0074] The signal separation step is in fact a step of un-mixing the transmitted plurality of signals **103**. For example, for the SIMO-BF-MAC channel, this step can be implemented by solving the following optimization problem (wherein {circumflex over (K)}.sub.l=K.sub.l).

[00002] $$\min_{\substack{s_l, .\mathrm{Math}. , s_{K_l} \in C \\ h_l, .\mathrm{Math}. , h_{K_l} \in C^{N_l}}} .\mathrm{Math}.\ Y_l - .\mathrm{Math}._{k=1}^{K_l} s_k h_{k,l}^T\ .\mathrm{Math}.$$

[0075] In the exchange step, the network devices **100** can then exchange two types of information: (i) the user ID (denoted as UID) of the separated user device signals **503** (second exchange information); or (ii) the index of the transmitted signal s in the constellation C (denoted as SIDX) of a separated user signal **504**, if the related user device **110** has been decoded at the first decoding step of the current interval **401**. The SIDX contains both the user ID and its payload and may be the first exchange information **104** described above.

[0076] The exchanged information may then be employed by the neighbor network devices **100** to improve their performance in the next decoding step or for the next interval **402**. Note that the exchange step may rely on the results of the second decoding step of the previous interval, and the first decoding step of the current interval **401**.

[0077] During the exchange step, a network device **100** transmits either UID or SIDX about their decoded user devices **110** to its neighbor network devices **100** according to the flow chart shown in FIG. **6**.

[0078] In particular, at block **601**, the network device **100** determines, whether a user device **110** has been decoded at the first decoding step of the current interval (here referred to as “resource block”). If yes, then in block **602** the network device **100** determines whether the user device **110** belongs to the second dataset **102** maintained at the network device **100**, i.e., whether a user ID of the user device **110** is in said second dataset. If yes, then at block **604**, the network device sends the first exchange information **104** to the neighboring network devices **100**, which are in the first dataset **101** and are associated with a second dataset **102** that includes the user ID. If no, then at block **605** the network device **100** sends the first exchange information **104** to the neighboring network devices **100**, which are in the first dataset **101** and are associated with a second dataset **102** that includes the user ID, and sends the second exchange information to the rest of the neighboring network devices **100** in the first dataset **101**.

[0079] If at block **601** the answer is no, then at block **603** the network device **100** determines whether the user device **110** has been decoded at the second decoding step of the previous interval. If yes, then at block **606** the network device sends the second exchange information to all neighboring network devices **100** in the first dataset **101**. If yes, then the network device **100** does nothing.

[0080] In the second decoding step, the network device **100** performs a second separation of the received plurality of signals **103**, while considering the previously decoded signals **503** (during the first decoding step) and the received SIDXs (first exchange information **104**) of decoded user devices **110** on neighboring network devices **100** (received in the exchange step). At a given network device **100**, let K′.sub.l<K.sub.l be the total number of successfully decoded user signals **503** coming from both the first decoding and exchanging step. For example, for the SIMO-BF-

MAC, the received signal Y.sub.l may be reformulated as shown in FIG. **7**(*a*).
[0081] FIG. **7**(*b*) notes as {circumflex over (K)}′.sub.l the number of active user devices **110** decoded, i.e., their signals **703** separated in the further separating, by the network device l by the end of the second decoding step.
[0082] An interference Cancellation (SIC) receiver **501** is an example of a decoder that can be applied here, since it uses the previously decoded signals to decode the unknown ones. In the case of the SIMO-BF-MAC, it can be implemented by solving the following optimization problem.

[00003] $$\min_{\substack{s_{K_l'+1},\ .\text{Math.}\ ,\ s_{K_l} \in C \\ h_l,\ .\text{Math.}\ ,\ h_{K_l} \in C^{N_l}}} .\text{Math.}\ Y_l - .\text{Math.}_{k'=1}^{K_l'}\ s_{k'}\, h_{k',l}^T + .\text{Math.}_{k=K_l'+1}^{K_l}\ s_k\, h_{k,l}^T\ .\text{Math.}$$

[0083] Note that this kind of receiver (interference cancellation) is employed in many detection schemes in the context of random access with joint activity detection and payload decoding.
[0084] In the update set, the lists of user IDs (the second datasets **102**) are updated based on the results of both first decoding step of the current interval **401**, and the second decoding of the previous interval.
[0085] In this update step, the lists of user IDs may be updated by keeping useful connections (to avoid oversized lists) and adding new ones following newly connected user devices **110**. The updating may be done based on the locally separated signals **503**, **703** and information received from neighboring network devices **100**. The connectivity graph may thus be updated in a distributed way, since each network device **100** performs the update process of its own subgraph (i.e., local update of the second datasets **102**). During this step, a network device **100** can: (i) add a user device **110** to the “local” network device lists of UIDs (second dataset **102** associated with the network device **100**) for a user device **110** whose signal **503**, **703** has been decoded locally, and tag the edge with a timestamp; (ii) add a user device **110** to a neighbor network device list of UIDs (second dataset **102** associated with a neighboring network device **100**), when the information has been transmitted during the exchange step, and tag the edge with a timestamp; (iii) prune elements of the lists of UIDs, for which the associated timestamp is older than a threshold (timeout).
[0086] The threshold may be set depending on the network conditions and can be updated by the upper layers of the network. For instance, the timeout threshold can be adjusted dynamically to have an optimized connectivity subgraph on each network device **100**. This prevents large lists of UIDs (useless entries) or undersized lists (insufficient amount of information to exchange).
[0087] In the following illustrative example, three APs as network devices **100** and five UEs as user devices **110** are considered. The example is presented throughout four time intervals (also referred to as four macro time shots). Within each interval, the timeline is represented following the four main steps of the proposed scheme described above. FIG. **8** summarizes the four intervals of the example.
[0088] In the first interval (see also FIG. **9**), the APs **1**, **2**, and **3** have successfully separated the following signals (s.sub.1, s.sub.3), (s.sub.4), and (s.sub.3), respectively. After the exchange step, no remaining signals are to be separated in the second decoding step, and the connectivity graph is updated accordingly; only new edges are added herein, (AP**1**-UE**1**), (AP**1**-UE**3**), (AP**3**-UE**3**), and (AP**2**-UE**4**).
[0089] In the second interval (see FIG. **10**), the APs **1**, **2**, and **3** have successfully separated the following signals (s.sub.1), (s.sub.2), and (s.sub.3, s.sub.4), respectively. After the exchange step, no remaining signals are to be separated in the second decoding step for APs **2** and **3**, but at AP**1**, the set (s.sub.1) still needs to be separated at the second decoding step. Then, the connectivity graph is updated accordingly; only new edges are added herein, (AP**1**-UE**5**), (AP**2**-UE**2**), and (AP**3**-UE**4**).
[0090] In the third interval (see FIG. **11**), the APs **1**, **2**, and **3** have successfully separated the following sets (s.sub.1, s.sub.3), (s.sub.4), and (s.sub.3), respectively. After the exchange step, no

remaining signals are to be separated in the second decoding step for the APs **1** and **2**, but at the AP **3**, the set (s.sub.5) still needs to be separated in the second decoding step. Herein, no new connection needs to be added to the connectivity graph.

[0091] In the fourth interval (see FIG. **12**), the APs **1** and **3** have successfully separated the following sets (s.sub.5) and (s.sub.3), respectively. Note here that the AP **3** transmits the information about the ID of UE **5** since it has been decoded during the second decoding step of the previous time interval. After the exchange step, no remaining signals are to be separated in the second decoding step. The connectivity graph is updated accordingly; the connection (AP**3**-UE**5**) is added accordingly and connections (AP**1**-UE**1**), (AP**1**-UE**3**), and (AP**3**-UE**3**) are removed because of timeout reach.

[0092] FIG. **13** shows a method **1300** for a network device **100** of a cell-free wireless communication network. The method **1300** may be performed by any network device **100** described above.

[0093] The method **1300** comprises a step **1301** of maintaining a first dataset **101** indicating one or more neighboring network devices **100**, and a step **1302** of maintaining a plurality of second datasets **102** each indicating one or more user IDs. One second dataset **102** of the plurality of second datasets is associated with the network device **100**, and each other second dataset **102** of the plurality of second datasets is associated with one of the one or more neighboring network devices **100** in the first dataset **101**.

[0094] The method **1300** further comprises a step **1303** of receiving a plurality of signals **103** from a plurality of user devices **110** in a first interval **401**, wherein each signal **503**, **703** comprises a respective user ID of one of the user devices **110** and a respective payload of one of the user devices **110**.

[0095] The method **1300** further comprises a step **1304** (corresponding to the first decoding step described above) of separating, in the first interval **401**, one or more signals **503** of the plurality of signals **103** to obtain the respective user IDs and the respective payloads of the one or more signals **503**.

[0096] The method **1300** further comprises a step **1305** (corresponding to the exchange step described above) sending, in the first interval **401**, first exchange information **104** to each neighboring network device **100**, which is in the first dataset **101** and is associated with a second dataset **102** that includes at least one of the obtained respective user IDs. The first exchange information **104** comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads. In the step **1305**, also first exchange information **104** from a neighboring network device **100** may be received, wherein the received first exchange information **104** comprises at least one of the respective user IDs and at least one of the respective payloads of the plurality of signals **103**.

[0097] The method **1300** may further comprise a step (corresponding to the second decoding step described above) of further separating, in the first interval **401**, one or more signals **703** of the plurality of signals **103** based on the previously separated signals **503** and the respective payloads in the received first exchange information **104**.

[0098] The method **1300** may further comprise a step (corresponding to the update step described above) of updating the second dataset **102** associated with the network device **100** based on the one or more separated signals **503** and the one or more further separated signals **703**, and/or updating one or more second datasets **102** associated with one or more neighboring network devices **100** based on first exchange information **104** received from the one or more neighboring network devices **100**.

[0099] In summary, this disclosure proposes a cooperation process in the context of random access, which includes structuring and updating datasets **101**, **102**, which indicate network device to user device connections (e.g., AP-UE connections) at each of a set of network devices **100**. This is done to exchange the separated signals **503**, **703** between the network devices **100** based on the learned local datasets **102**.

[0100] The present disclosure has been described in conjunction with various embodiments as examples as well as implementations. However, other variations can be understood and effected by those persons skilled in the art and practicing the claimed matter, from the studies of the drawings, this disclosure and the independent claims. In the claims as well as in the description the word "comprising" does not exclude other elements or steps and the indefinite article "a" or "an" does not exclude a plurality. A single element or other unit may fulfill the functions of several entities or items recited in the claims. The mere fact that certain measures are recited in the mutual different dependent claims does not indicate that a combination of these measures cannot be used in an advantageous implementation.

Claims

1. A network device (**100**) for a cell-free wireless communication network, the network device (**100**) being configured to: maintain a first dataset (**101**) indicating one or more neighboring network devices (**100**); maintain a plurality of second datasets (**102**) each indicating one or more user IDs, wherein one second dataset (**102**) of the plurality of second datasets is associated with the network device (**100**), and each other second dataset (**102**) of the plurality of second datasets is associated with one of the one or more neighboring network devices (**100**) in the first dataset (**101**); receive a plurality of signals (**103**) from a plurality of user devices (**110**) in a first interval (**401**), wherein each signal (**503**, **703**) comprises a respective user ID of one of the user devices (**110**) and a respective payload of one of the user devices (**110**); separate, in the first interval (**401**), one or more signals (**503**) of the plurality of signals (**103**) to obtain the respective user IDs and the respective payloads of the one or more signals (**503**); and send, in the first interval (**401**), first exchange information (**104**) to each neighboring network device (**100**), which is in the first dataset (**101**) and is associated with a second dataset (**102**) that includes at least one of the obtained respective user IDs; wherein the sent first exchange information (**104**) comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.

2. The network device (**100**) according to claim 1, wherein, if a particular user ID of the obtained respective user IDs is not in the second dataset (**102**) associated with the network device (**100**), the network device (**100**) is further configured to: send, in the first interval (**401**), second exchange information to each neighboring network device (**100**), which is in the first dataset (**101**) and is associated with a second dataset (**102**) that does not include the particular user ID; wherein the second exchange information comprises the particular user ID but not the respective payload corresponding to the particular user ID.

3. The network device (**100**) according to claim 1, further configured to: receive, in the first interval (**401**), first exchange information (**104**) from a neighboring network device (**100**), wherein the received first exchange information (**104**) comprises at least one of the respective user IDs and at least one of the respective payloads of the plurality of signals (**103**); and further separate, in the first interval (**401**), one or more of the plurality of signals (**103**) based on the separated signals (**503**) and the respective payloads in the received first exchange information (**104**).

4. The network device (**100**) according to claim 2, further configured to: send, in a second interval (**402**), the second exchange information to each neighboring network device (**100**) in the first dataset (**101**).

5. The network device (**100**) according to claim 1, configured to: perform the separation of the one or more signals (**503**) of the plurality of signals (**103**), in the first interval (**401**), without prior knowledge of the respective user IDs and the respective payloads of the one or more signals (**503**).

6. The network device (**100**) according to claim 4, wherein the first exchange information (**104**) received from the neighboring network device (**100**) comprises one or more respective user IDs and one or more respective payloads that are different than in the sent first exchange information (**104**).

7. The network device (**100**) according to claim 1, wherein in the further separation of the one or more signals (**703**) of the plurality of signals (**103**), in the first interval (**401**), the network device (**100**) is configured to perform interference cancellation.
8. The network device (**100**) according to claim 3, further configured to: update the second dataset (**102**) associated with the network device (**100**) based on the one or more separated signals (**503**) and the one or more further separated signals (**703**), and/or update one or more second datasets (**102**) associated with one or more neighboring network devices (**100**) based on the first exchange information (**104**) received from the one or more neighboring network devices (**100**).
9. The network device (**100**) according to claim 8, configured to update the second dataset (**102**) associated with the network device (**100**) and/or the one or more second datasets (**102**) associated with the one or more neighboring network devices (**100**) by at least one of: adding the one or more user IDs to the second dataset (**102**) associated with the network device (**100**); and adding the one or more user IDs to one or more second datasets (**102**) associated with one or more neighboring network devices (**100**); and removing one or more user IDs from the second dataset (**102**) associated with the network device (**100**); and removing one or more user IDs from the one or more second datasets (**102**) associated with the one or more neighboring network devices (**100**).
10. The network device (**100**) according to claim 9, further configured to associate a timestamp with any user ID added to a second dataset (**102**).
11. The network device (**100**) according to claim 9, further configured to remove any user ID from the second dataset (**102**), if a timestamp associated with the user ID is older than a threshold.
12. The network device (**100**) according to claim 11, further configured to adjust the threshold based on at least one of a network condition and an upper layer configuration.
13. The network device (**100**) according to claim 9, further configured to: add a user ID of a separated signal or a further separated signal to the second dataset (**102**) of the network device (**100**), if the user ID of the separated signal is not already in the second dataset (**102**) of the network device (**100**); and/or add a user ID in first exchange information (**104**) received from a neighboring network device (**100**) to the second dataset (**102**) of that neighboring network device (**100**), if the user ID in the sent first exchange information (**104**) is not already in the second dataset (**102**) of that neighboring network device (**100**).
14. A method (**1300**) for a network device (**100**) of a cell-free wireless communication network, the method (**1300**) comprising: maintaining (**1301**) a first dataset (**101**) indicating one or more neighboring network devices (**100**); maintaining (**1302**) a plurality of second datasets (**102**) each indicating one or more user IDs, wherein one second dataset (**102**) of the plurality of second datasets is associated with the network device (**100**), and each other second dataset (**102**) of the plurality of second datasets is associated with one of the one or more neighboring network devices (**100**) in the first dataset (**101**); receiving (**1303**) a plurality of signals (**103**) from a plurality of user devices (**110**) in a first interval (**401**), wherein each signal (**503**, **703**) comprises a respective user ID of one of the user devices (**110**) and a respective payload of one of the user devices (**110**); separating (**1304**), in the first interval (**401**), one or more signals (**503**) of the plurality of signals to obtain the respective user IDs and the respective payloads of the one or more signals (**103**); and sending (**1305**), in the first interval (**401**), first exchange information (**104**) to each neighboring network device (**100**), which is in the first dataset (**101**) and is associated with a second dataset (**102**) that includes at least one of the obtained respective user IDs; wherein the first exchange information (**104**) comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.
15. A computer program comprising instructions which, when the program is executed by a computer, cause the computer to perform a method (**1300**), the method (**1300**) comprising: maintaining (**1301**) a first dataset (**101**) indicating one or more neighboring network devices (**100**); maintaining (**1302**) a plurality of second datasets (**102**) each indicating one or more user IDs, wherein one second dataset (**102**) of the plurality of second datasets is associated with the network

device (**100**), and each other second dataset (**102**) of the plurality of second datasets is associated with one of the one or more neighboring network devices (**100**) in the first dataset (**101**); receiving (**1303**) a plurality of signals (**103**) from a plurality of user devices (**110**) in a first interval (**401**), wherein each signal (**503**, **703**) comprises a respective user ID of one of the user devices (**110**) and a respective payload of one of the user devices (**110**); separating (**1304**), in the first interval (**401**), one or more signals (**503**) of the plurality of signals to obtain the respective user IDs and the respective payloads of the one or more signals (**103**); and sending (**1305**), in the first interval (**401**), first exchange information (**104**) to each neighboring network device (**100**), which is in the first dataset (**101**) and is associated with a second dataset (**102**) that includes at least one of the obtained respective user IDs; wherein the first exchange information (**104**) comprises at least one of the obtained respective user IDs and at least one of the obtained respective payloads.